

ATLAS stability atlas: categorical operating-limit view

estimator	PLL	fragile 0.05	fragile 0.10	fragile 3	fragile 1	fragile 1	fragile 0.50	fragile 1e-04	fragile 0.10	fragile 0.10
	SOGI-PLL	fragile 0.50	fragile 10	fragile 5	fragile 25	fragile 8	fragile 0.50	fragile 0.03	usable 10	usable 8
	SOGI-FLL	fragile 0.50	fragile 3	fragile 5	usable 75	fragile 8	fragile 2	usable 0.10	stable 10+	fragile 3
	T3-SOGI-PLL	fragile 0.05	fragile 0.10	fragile 5	fragile 1	fragile 1	fragile 0.50	fragile 1e-04	fragile 0.10	fragile 0.10
	ZCD	fragile 0.50	fragile 5	fragile 5	usable 90	usable 20	fragile 1	fragile 0.01	stable 10+	usable 5
	IPDFT	fragile 0.50	fragile 5	fragile 5	fragile 35	fragile 5	fragile 1	fragile 0.03	stable 10+	usable 5
	TFT	fragile 0.50	fragile 5	fragile 5	usable 50	fragile 10	fragile 1	fragile 0.03	stable 10+	usable 5
	ESPRIT	fragile 1	fragile 10	fragile 5	fragile 20	fragile 2	fragile 0.50	fragile 0.03	stable 10+	usable 8
	Prony	invalid flag	invalid flag	invalid flag	invalid flag	fragile 1	fragile 0.50	invalid flag	fragile 0.10	fragile 0.10
	MUSIC	fragile 1	fragile 10	fragile 5	fragile 20	fragile 2	fragile 0.50	fragile 0.03	stable 10+	usable 8
	EKF	fragile 0.20	fragile 0.50	usable 30	stable 90+	usable 15	fragile 8	usable 0.10	stable 10+	fragile 0.10
	UKF	fragile 0.50	fragile 3	fragile 10	usable 75	usable 15	fragile 3	usable 0.10	stable 10+	fragile 2
	RA-EKF	fragile 0.50	fragile 3	fragile 10	stable 90+	fragile 5	fragile 2	usable 0.10	stable 10+	fragile 2
	LKF	fragile 0.05	fragile 0.10	fragile 5	fragile 1	fragile 1	fragile 0.50	fragile 1e-04	fragile 0.10	fragile 0.10
	LKF2	fragile 0.20	fragile 0.50	fragile 10	usable 50	fragile 2	fragile 2	fragile 0.03	stable 10+	fragile 0.10
	RLS	fragile 0.05	fragile 0.10	invalid flag	invalid flag	invalid flag	invalid flag	invalid flag	fragile 0.10	fragile 0.10
	TKEO	fragile 0.05	fragile 0.10	fragile 5	fragile 1	fragile 1	fragile 1	fragile 1e-04	fragile 0.10	fragile 0.10
	Koopman	fragile 1	fragile 10	fragile 5	fragile 35	fragile 3	fragile 1	fragile 0.03	stable 10+	usable 5
		Freq step	RoCoF	Phase jump	Mag step	Harmonics	Interharm.	Noise	AM	FM
		stress family								
		stable: RMSE <= 0.05 Hz across tested range				fragile: threshold crossed early				
		usable: threshold reached after >=50% of max severity				invalid/diverged: non-finite, invalid-rate, or bound-hit flag				

Cell value is the first severity where frequency RMSE exceeds 0.05 Hz, in each stress family's units; '+' means the guide was not exceeded. ATLAS artifact: 17 canonical estimators plus MUSIC extra; PI-GRU is handled separately.